The empirical recurrence for OEIS A214108, proved by transfer matrix

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Abstract

OEIS A214108 counts the proper colourings of a grid $n \times 5$, wrapped in the direction of width 5, with at most 4 colours and counted up to renaming the colours, and carries an empirical recurrence of order 2 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. Counting up to renaming is a fixed rational combination of ordinary colouring counts; each of those is a walk count; and because permuting the colours preserves everything in sight, the walk may be run on the colour-patterns rather than on the colourings, which is what brings the state count down to $S = 270$ and makes the exact residual test affordable.

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1 The conjecture

OEIS A214108 is "Number of 0..3 colorings on an nX5 array circular in the 5 direction with new values 0..3 introduced in row major order.". It has offset 1 and begins

$$10, 670, 44900, 3008980, 201647240, 13513419640, 905603817680, 60689173906000, \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: a(n) = 68*a(n-1) -66*a(n-2).

As of the "Last modified" line on the live entry (Oct 03 2025 16:51:43, revision 10) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

A *colouring* here is proper: cells adjacent horizontally or vertically carry different values, and "circular in one direction" means the two ends of that direction are adjacent as well. The check that this is the right reading is the entry's own first term for the one-row case: a cycle of W cells has $(i-1)^W + (-1)^W(i-1)$ proper colourings over i colours, which at $W = 6$, $i = 3$ is 66, and $66/3! = 11$ is what the corresponding entry of this family publishes.

The clause "new values 0..3 introduced in row major order" says the array is the canonical representative of its equality pattern, so what is counted is patterns, not arrays. Write N_j for the number of admissible patterns using exactly j colours and L_i for the number of admissible colourings over an alphabet of i colours. A colouring over i colours is a pattern together with an injection of its colours into the alphabet, so

$$L_i \;=\; \sum_j N_j \, i(i-1)\cdots(i-j+1),$$

a triangular system with unit diagonal, hence invertible over $\mathbf{Q}$. The entry counts $a = N_1 + \cdots + N_4$, so

$$a \;=\; \sum_{i=1}^{4} w_i \, L_i$$

for one fixed rational vector $w = \mathbf{1}^\top F^{-1}$, F the matrix of falling factorials. Being properly coloured is a statement about equalities alone, so it is invariant under permuting the colours and the reduction is legitimate.

3 Each L_i is a walk count

A row of the array is a proper colouring of the cycle C_5, since the wrapping makes its first and last cells adjacent, and two consecutive rows must differ in every one of the 5 positions. So $L_i(n)$ is the number of walks of $n-1$ steps in the digraph whose vertices are the proper colourings of C_5 over i colours and whose edges join colourings differing everywhere:

$$L_i(n) \;=\; \iota^\top M_i^{\,n-1} \iota, \qquad \iota = (1, \ldots, 1)^\top.$$

There are $(i-1)^5 + (-1)^5(i-1)$ such colourings, which at $i = 4$ is already more than is comfortable; Lemma 1 removes the difficulty.

Lemma 1. *Let a group G act on the vertices of a digraph preserving its edges, and let v be constant on the G-orbits. Then Mv is constant on the G-orbits, and for a representative s of the orbit O,*

$$(Mv)(O) \;=\; \sum_{O'} A_{OO'} \, v(O'), \qquad A_{OO'} = \#\{t \in O' : \; s \to t\}.$$

So an iteration started at a G-invariant vector may be carried out on the orbits.

Proof. $\sigma \in G$ is an edge-preserving bijection, so it carries the out-neighbours of s onto those of σs; with v constant on orbits, $(Mv)(s)$ and $(Mv)(\sigma s)$ are the same sum. The count $A_{OO'}$ is likewise independent of which representative of O is used. □

Both endpoint vectors above are invariant under the relevant group — the all-ones vectors under all of S_i, and, for a diagonal entry $(T_i^m)_{ss}$, the vector e_s under the subgroup fixing s — so the whole computation runs on orbits. That is the entire reason the state count is $S = 270$ rather than the number of colourings.

Assembling the pieces, $a(n)$ is $\iota^\top M^j \tau$ on the disjoint union of the orbit digraphs for $i = 1, \ldots, 4$, with ι carrying the weights w_i cleared of denominators; the walk index j and the entry's n differ by a constant, read off by matching the entry's published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 2. *Let $q(t) = t^2 - \sum_i c_i t^{2-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+2} - \sum_i c_i M^{j+2-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 3.

$$a(n) \;=\; 68\,a(n-1) - 66\,a(n-2)$$

for every $n > 2$.

Proof. The vector $w = q(M)\tau$ was formed by 2 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots$, again exactly; those integers vanish from the index corresponding to $n = 2 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. The rational weights w_i are cleared by a common denominator before any of this, so the whole computation is over the integers. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.